Software Requirements Specification (SRS)

OneTap365 Mobile Application

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the OneTap365 mobile application - a classified listing marketplace platform available on Android (Play Store) and iOS (App Store).

1.2 Scope

The OneTap365 mobile application will provide a comprehensive platform for buying, selling, and discovering classified listings across multiple categories including Properties, Electronics, Vehicles, Jobs, and Services. The system will support multiple user roles with varying access levels and include membership plans for enhanced features.

1.3 Technology Stack

- **Frontend**: Flutter (Cross-platform for iOS & Android)
- **Backend**: Node.js with Express.js
- **Database**: MongoDB
- **Authentication**: JWT-based authentication
- **Storage**: AWS S3 / Cloud Storage for images and documents
- **Push Notifications**: Firebase Cloud Messaging (FCM)
- **Payment Gateway**: Razorpay / Stripe / PayU

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2. System Architecture

2.1 User Roles & Access Levels

2.1.1 Master Admin Panel

Access Level: Full System Control

Features & Permissions:

- Complete dashboard with analytics and reports
- User management (all roles)
- Vendor approval and management
- Employee and manager management
- Category and subcategory management
- Listing moderation and approval
- Payment and transaction management
- Membership plan configuration
- Commission settings
- App settings and configurations
- Push notification management
- Reporting and analytics
- Helpline management
- Content management (blogs, banners)
- Review and rating moderation
- System logs and audit trails

2.1.2 Vendor Panel

Access Level: Own Listings & Business Management

Features & Permissions:

- Vendor dashboard with personal analytics
- Create, edit, delete own listings
- Listing management (active, pending, expired)
- View listing statistics (views, clicks, inquiries)
- Membership plan subscription
- Payment history
- Lead management (inquiries received)
- Profile management
- Upload documents for verification
- Performance reports
- Customer inquiries and chat
- Featured listing promotion
- Bulk upload functionality (CSV)

2.1.3 Manager Panel

Access Level: Operational Management

Features & Permissions:

- Manager dashboard with team analytics
- Oversee assigned employees
- Listing verification and moderation
- Customer support management
- Generate reports for admin
- Handle escalated queries
- Monitor vendor activities
- Approve/reject listings (delegated)
- View analytics and performance metrics
- Communication with vendors and employees

2.1.4 Employee Panel

Access Level: Operational Support

Features & Permissions:

- Employee dashboard
- Listing verification (basic)
- Customer support tickets
- Vendor query resolution
- Data entry and management
- Report to manager
- View assigned tasks
- Basic analytics view
- Communication tools

2.1.5 User Panel (Customer)

Access Level: Browse, Buy, and Sell

Features & Permissions:

- User dashboard
- Browse and search listings
- Post free/paid listings (based on membership)
- Save favorite listings
- Contact vendors/sellers
- In-app chat with sellers
- Profile management
- Purchase history
- Membership subscription
- Review and rating
- Inquiry history
- Wishlist/Saved searches
- Notifications preferences

- Location-based search

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3. Functional Requirements

3.1 User Authentication & Authorization

3.1.1 Registration

- Phone number verification (OTP)
- Email verification
- Social login (Google, Facebook, Apple)
- Role selection during registration
- Profile creation with KYC for vendors
- Terms and conditions acceptance

3.1.2 Login

- Phone/Email + Password
- OTP-based login
- Biometric authentication (Fingerprint/Face ID)
- Remember me functionality
- Forgot password with OTP recovery

3.1.3 Profile Management

- Edit personal information
- Upload profile picture
- Change password
- Manage contact details
- Notification settings
- Language preferences
- Delete account option

3.2 Master Admin Features

3.2.1 Dashboard

- Total users, vendors, employees count
- Active listings statistics
- Revenue analytics (daily, weekly, monthly)
- Membership subscriptions overview

- Recent activities feed
- Pending approvals count
- System health monitoring

3.2.2 User Management

- View all users with filters (role, status, date)
- Search users
- Activate/deactivate accounts
- Delete users
- View user details and activity logs
- Send notifications to specific users/groups
- Export user data

3.2.3 Vendor Management

- Pending vendor approval queue
- Approve/reject vendor registration
- View vendor documents (KYC)
- Vendor performance metrics
- Commission management per vendor
- Suspend/ban vendors
- Vendor tier management

3.2.4 Category Management

- Create main categories
- Create subcategories
- Edit/delete categories
- Set category icons and images
- Category ordering
- Enable/disable categories

3.2.5 Listing Management

- View all listings (filter by status, category, date)
- Approve/reject listings
- Feature listings
- Mark listings as verified
- Delete inappropriate listings
- Bulk actions
- Listing reporting system

3.2.6 Membership Plan Management

- Create membership plans

- Set pricing and duration
- Define features per plan
- Free vs Paid listing limits
- Featured listing allowances
- Discount management
- Plan activation/deactivation

3.2.7 Payment & Transaction Management

- View all transactions
- Payment gateway settings
- Refund management
- Revenue reports
- Commission tracking
- Invoice generation
- Payment disputes

3.2.8 Content Management

- Create/edit blog posts
- Manage banners and advertisements
- Homepage content management
- Notification templates
- Email templates
- Static page management (About, Terms, Privacy)

3.2.9 Service Management

- Add service categories
- Manage service providers
- Service scheduling system
- Service pricing management

3.2.10 Analytics & Reports

- User growth reports
- Revenue reports
- Listing performance
- Category-wise analytics
- Geographic analytics
- Custom date range reports
- Export reports (PDF, Excel)

3.3 Vendor Panel Features

3.3.1 Dashboard

- Active listings count
- Total views and clicks
- Inquiries received
- Membership status
- Quick actions (create listing)
- Performance graphs

3.3.2 Listing Management

- Create new listing with form:
    - Title, description
    - Category and subcategory selection
    - Price
    - Multiple images (up to 10)
    - Location (auto-detect + manual)
    - Contact details
    - Additional attributes based on category
- Edit existing listings
- Delete listings
- Mark as sold/unavailable
- Duplicate listing
- Listing preview before publish
- Schedule listing publication
- Bulk upload via CSV

3.3.3 Inquiry Management

- View all inquiries
- Filter by date, listing
- Respond to inquiries
- Mark as resolved
- Priority marking
- Export inquiries

3.3.4 Membership Management

- View current plan
- Upgrade/downgrade plan
- Subscription history
- Payment invoices
- Auto-renewal settings
- Feature usage tracking

3.3.5 Analytics

- Listing-wise views and clicks
- Inquiry conversion rate
- Best performing listings
- Traffic sources
- Time-based analytics

3.4 Manager Panel Features

3.4.1 Dashboard

- Team performance overview
- Pending tasks
- Escalated issues
- Today's assignments
- Quick stats

3.4.2 Team Management

- View assigned employees
- Assign tasks to employees
- Monitor employee performance
- Approve employee actions

3.4.3 Listing Moderation

- Review pending listings
- Approve/reject with comments
- Quality check
- Category verification

3.4.4 Customer Support

- View support tickets
- Assign to employees
- Escalate to admin
- Resolve queries
- Response templates

3.4.5 Reports

- Generate team reports
- Vendor activity reports
- Issue resolution reports
- Submit to admin

3.5 Employee Panel Features

3.5.1 Dashboard

- Assigned tasks
- Pending approvals
- Today's targets
- Performance metrics

3.5.2 Task Management

- View assigned tasks
- Update task status
- Add notes/comments
- Mark as completed

3.5.3 Listing Verification

- Verify listing details
- Check for policy violations
- Flag inappropriate content
- Submit for manager approval

3.5.4 Customer Support

- Handle support tickets
- Respond to queries
- Use predefined responses
- Escalate complex issues

3.6 User (Customer) Panel Features

3.6.1 Home Screen

- Search bar (keyword, location)
- Category tiles
- Featured listings carousel
- Popular listings
- Recent listings
- Banner advertisements
- Quick filters

3.6.2 Browse & Search

- Category-wise browsing
- Advanced filters:
    - Price range
    - Location
    - Sort by (newest, price, relevance)
    - Condition (new/used)
    - Category-specific filters
- Search suggestions
- Save search preferences
- Location-based listings

3.6.3 Listing Details View

- Image gallery with zoom
- Title and description
- Price
- Seller information
- Location on map
- Contact options (call, WhatsApp, chat)
- Share listing
- Report listing
- Save to favorites
- Similar listings
- Seller's other listings

3.6.4 Post Listing

- Same as vendor with plan restrictions
- Free listings (limited)
- Paid/featured listings
- Draft saving
- Form validation
- Image upload (compression)

3.6.5 My Listings

- View posted listings
- Edit/delete
- Mark as sold
- View inquiries
- Boost listing (paid feature)
- Listing statistics

3.6.6 Favorites/Wishlist

- Save listings
- Organize in folders
- Share wishlist
- Get price drop alerts

3.6.7 Inquiries & Messages

- View sent inquiries
- In-app chat with sellers
- Call history
- Message templates
- Block users
- Notification for responses

3.6.8 Profile

- Edit profile
- Verification badge (KYC)
- Reviews and ratings
- Member since date
- Total listings

3.6.9 Membership

- View available plans
- Subscribe to plans
- Payment integration
- Manage subscription
- Feature comparison

3.6.10 Service Booking

- Browse services (AC, cleaning, etc.)
- Select service type
- Choose date and time slot
- Enter address
- Book appointment
- View booking history
- Track service status
- Rate service

3.7 Common Features (All Roles)

3.7.1 Notifications

- Push notifications
- In-app notifications
- Email notifications
- SMS notifications
- Notification preferences
- Notification history

3.7.2 Chat System

- Real-time messaging
- Image sharing in chat
- Online/offline status
- Read receipts
- Block/report users
- Chat history

3.7.3 Reviews & Ratings

- Rate sellers/vendors
- Write reviews
- View received reviews
- Report inappropriate reviews
- Star rating system (1-5)

3.7.4 Location Services

- GPS location detection
- Manual location entry
- Location-based search
- Map integration (Google Maps)
- Nearby listings

3.7.5 Settings

- Account settings
- Privacy settings
- Notification preferences
- Language selection
- Theme (light/dark mode)
- Help & support
- About app
- Terms & privacy policy

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4. Membership Plans

4.1 Plan Structure

4.1.1 Free Plan (Default)

Features:

- Browse unlimited listings
- Post up to 3 listings per month
- Basic search and filters
- Contact sellers
- Standard support

Limitations:

- No featured listings
- No priority in search results
- Ads displayed
- Limited image uploads (3 per listing)

4.1.2 Basic Plan

Price: ₹299/month or ₹2,999/year

Features:

- Post up to 15 listings per month
- 1 featured listing per month
- 5 images per listing
- Basic analytics
- Priority support
- Reduced ads

Limitations:

- No top placement
- Limited boost options

4.1.3 Premium Plan

Price: ₹699/month or ₹6,999/year

Features:

- Unlimited listings

- 5 featured listings per month
- 10 images per listing
- Advanced analytics
- Priority support (24/7)
- No ads
- Listing boost (5 per month)
- Verification badge
- Bulk upload

Limitations:

- Standard search placement

4.1.4 Enterprise Plan

Price: ₹1,999/month or ₹19,999/year

Features:

- Everything in Premium
- Unlimited featured listings
- Top search placement
- Dedicated account manager
- API access
- Custom branding
- Featured vendor badge
- Priority listing approval
- Custom reports
- Social media promotion

4.2 Plan Management

- Plan comparison table
- Upgrade/downgrade anytime
- Prorated billing
- Auto-renewal
- Cancel anytime
- Trial period (7 days for Premium)
- Discount codes
- Referral benefits

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5. Categories & Subcategories

5.1 Properties

- Apartments & Flats
- Houses
- Plots & Land
- Commercial Properties
- PG & Hostels
- Rent/Sale options

5.2 Business & Industry

- Business for Sale
- Industrial Equipment
- Office Equipment
- Commercial Vehicles

5.3 Cars & Vehicles

- Cars
- Bikes
- Scooters
- Commercial Vehicles
- Spare Parts
- Accessories

5.4 Electronics

- Mobile Phones
- Laptops & Computers
- TVs & Appliances
- Cameras
- Other Electronics

5.5 Food & Agriculture

- Fresh Produce
- Seeds & Plants
- Farm Equipment
- Livestock

5.6 Health & Beauty

- Health Products
- Beauty Products

- Fitness Equipment
- Salon Services

5.7 Hobby, Sport & Kids

- Sports Equipment
- Toys & Games
- Books
- Musical Instruments

5.8 Home Appliances

- Kitchen Appliances
- Washing Machines
- Refrigerators
- Air Conditioners
- Home Décor

5.9 Jobs

- Full-time
- Part-time
- Freelance
- Internships
- Work from Home

5.10 Services

- AC Installation/Repair
- Home Cleaning
- Painting & Interior
- Pest Control
- Plumber
- Electrician
- Carpenter
- Property Management

5.11 Others

- Miscellaneous items

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6. Service Booking System

6.1 Service Categories

- AC (Air Conditioner) - Installation/Repair
- Home Cleaning Services
- Electronics & Appliances Repair
- Painting, Interior, and Civil Work
- Pest Control
- Plumber Services
- Electrician Services
- Carpenter Services
- Property Management
- Property Interior Designing
- Property Documentation
- Property Construction

6.2 Booking Flow

1. User selects service category
2. Choose service type (installation/repair/etc.)
3. Select date and time slot
4. Enter service address
5. Add special instructions
6. Review and confirm booking
7. Payment (online/COD)
8. Booking confirmation

6.3 Service Provider Features

- Accept/reject bookings
- Update service status
- Complete booking
- Upload before/after photos
- Generate invoice
- Customer communication

6.4 Booking Management

- View upcoming bookings
- Booking history
- Reschedule/cancel
- Track service person
- Rate service
- Request callback

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7. Investment Calculator Feature

7.1 Calculator Inputs

- Monthly Investment Amount (₹)
- Rate of Interest (ROI %)
- Time Period (Years)

7.2 Calculator Outputs

- Total Invested Amount
- Estimated Returns
- Total Value
- Visual representation (graph/chart)

7.3 Features

- Save calculations
- Compare multiple scenarios
- Share results
- Investment advice/tips
- Connect with property advisors

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8. Non-Functional Requirements

8.1 Performance

- App load time: < 3 seconds
- API response time: < 2 seconds
- Image loading: Progressive/lazy loading
- Smooth scrolling (60 FPS)
- Offline mode for browsing cached data
- Database indexing for fast queries

8.2 Security

- HTTPS/SSL encryption
- JWT token-based authentication
- Password encryption (bcrypt)

- Data validation and sanitization
- SQL injection prevention
- XSS protection
- Rate limiting on APIs
- Two-factor authentication (optional)
- Secure payment gateway integration
- GDPR compliance
- Data backup and recovery

8.3 Scalability

- Horizontal scaling capability
- Load balancing
- CDN for image delivery
- Caching mechanism (Redis)
- Database sharding for large datasets
- Microservices architecture (future)

8.4 Usability

- Intuitive UI/UX
- Material Design (Android) / Human Interface Guidelines (iOS)
- Multi-language support (English, Hindi)
- Accessibility features
- Onboarding tutorial for new users
- Help and documentation
- In-app support chat

8.5 Compatibility

- Android: Version 6.0 (API 23) and above
- iOS: Version 12.0 and above
- Responsive design for tablets
- Support for various screen sizes

8.6 Reliability

- 99.9% uptime
- Error handling and logging
- Crash reporting (Firebase Crashlytics)
- Automated testing
- Regular backups